

Solution Requirements

Date	8 July 2026
Team ID	XXXXXX
Project Name	Smart Lender
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- Functional Requirements (FR) define specific behaviors, functions, or features of the system (What the system should *do*).
- Non-Functional Requirements (NFR) specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	The web application provides secure dashboard session routing for credit officers and financial analysts to restrict predictor engine access to authorized personnel.	Medium
2	Authorization levels	Credit Officers / Analysts: Full access to input individual parameters, trigger batch row processing, and review high-risk default flags. Applicants: Read/write permissions strictly limited to	High

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
		submitting their own onboarding fields via the dynamic web form.	
3	External interfaces	Connects natively to an active Flask web server application instance deployed directly onto the IBM Cloud architecture.	High
4	Transactions processing	Ingests structured multi-attribute customer profiles (Gender, Marital Status, Education, Income, Loan Amount, Credit History) submitted through UI forms and transforms them into standard numerical evaluation arrays.	High
5	Reporting	Generates real-time visual assessment windows on the dashboard, displaying classification output results instantly as either a green-lit Fast-Track Approval or a red-flagged High-Risk Verification Alert.	High
6	Business rules, etc.	Follows precise financial data cleaning rules: missing continuous values are automatically resolved via mean imputation, missing categorical inputs are completed using mode imputation, and text variables undergo binary label encoding.	High
7	Compliance to laws or regulations	Adheres strictly to institutional credit-risk mitigation frameworks to actively minimize non-performing assets through automated, compliance-backed data screening.	Medium

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
8	Other	Embeds a production-ready, highly tuned XGBoost Classifier Model that executes predictive inference leveraging its validated \$94.7\%\$ training accuracy and \$81.1\%\$ testing accuracy threshold.	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system must ingest raw web inputs, execute missing value imputations, run XGBoost model inference, and output a result rapidly.	Real-time prediction results render on-screen in < 2 seconds.
2	Scalability	The backend configuration must scale effectively to process high volumes of simultaneous queries from multiple risk assessment analysts.	Supports concurrent batch processing of up to 1,000 applicant rows during peak validation hours.
3	Security & Data Privacy	All inputs submitted by banking users and personal candidate data must be kept isolated inside the runtime session.	Secured data parameters handling via HTTPS communication over IBM Cloud.
4	Reliability & Availability	The deployed Flask microservice instance must keep running smoothly to provide loan decisions	Target of 99.9% application uptime during standard banking hours.

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
		whenever accessed by credit officers.	
5	Usability & Accessibility	Form layouts, text sizes, drop-down selections, and alert fields must stay straightforward and completely intuitive for non-technical credit personnel.	Clear visual color coding: Green alerts for Low-Risk Fast-Track, Red alerts for High-Risk Verification.
6	Other	Maintainability: Modular separation between data preprocessing modules, the stored serialized model file (.pkl), and the user-facing web routing scripts.	Clear code structures allowing replacement or retuning of models in < 1 hour of downtime.